Context-Aware PPE and Worker–Machinery Proximity Monitoring

Object Detection, Tracking and Temporal Risk Scoring

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Deep Learning Project Proposal

Problem Statement & Motivation

Problem statement

Existing camera-based safety systems detect workers, machinery, helmets and vests as separate objects in individual frames. A helmet appearing in the frame does not prove that every worker is wearing one, and a person near a machine is not automatically in danger. These systems often lack worker-level PPE association, cabin-operator context and persistent proximity analysis across video frames.

- PPE-only alerts ignore machinery motion and worker location.
- A person visible inside a crane or excavator cabin may trigger a false proximity warning.
- A brief zone crossing and prolonged exposure should not receive the same risk level.
- Repeated false alarms reduce trust and can cause safety staff to ignore genuine warnings.

Project motivation

Convert frame-level detections into worker-specific, explainable safety events that use visual proximity, machinery motion, PPE status and exposure duration.

Who is affected?

- **Construction workers:** face direct exposure to moving machinery and may also work without correctly detected PPE.
- **Machine operators:** can trigger false proximity alarms when a camera sees them inside an excavator, crane or other cabin.
- **Site supervisors and safety officers:** must inspect repeated alerts and can suffer alert fatigue when warnings lack context.
- **Visitors and subcontractors:** may enter restricted work areas without understanding local machinery hazards.
- **Contractors and project owners:** carry operational, legal and financial consequences when preventable incidents occur.
- **Workers' families and nearby communities:** bear the long-term human impact of serious workplace injuries and fatalities.

Direct user of the system

The immediate user is the site safety team, which receives worker-specific, explainable alerts instead of raw detection boxes.

Why Deep Learning?

Tasks that require learned visual features

- Detect workers across scale, pose and partial occlusion
- Recognise visually different machinery categories
- Detect small helmets and safety vests under changing illumination
- Generalise beyond fixed colour and shape thresholds

Tasks handled by explicit system logic

- ByteTrack maintains temporary video identities
- Geometry associates PPE with worker regions
- Trajectories estimate machinery motion and exposure duration
- Temporal rules calculate and stabilise risk alerts

Scope distinction

Deep learning performs visual detection. Tracking, association and risk scoring form a transparent decision layer rather than another unverified neural model.

Practical / Technical Relevance

Practical relevance

- Works with recorded footage or existing monocular CCTV feeds
- Provides the reason and duration behind every warning
- Reduces cabin-operator and one-frame false alarms
- Creates structured event logs for review and audit
- Supports human safety staff; it does not replace them

Technical relevance

- Two specialised detectors trained from random initialisation
- Frame-level fusion followed by multi-object tracking
- Worker-specific PPE association with temporal voting
- Motion-adaptive danger zones in image space
- Event-level evaluation against a fixed-zone baseline

Background / Problem Context

- Published construction-safety work has established that modern object detectors can recognise workers, helmets, vests and machinery.
- Most benchmarks still score individual images using precision, recall and mean average precision.
- A site-safety decision requires additional context: which PPE belongs to which person, whether machinery is moving, and how long the worker remains exposed.
- A visible person near a machine is not always unsafe. A confirmed operator inside a cabin must be treated differently from an external pedestrian.
- A short accidental detection should not create the same alert as a persistent worker–machinery interaction.
- SafeVision therefore focuses on the transition from frame-level object detection to worker-specific temporal risk events.

The actual problem being solved

Input

- Recorded or live monocular construction-site video
- Multiple workers and machines
- Occlusion, small PPE objects and changing illumination

Required reasoning

- Detect workers, relevant machinery and PPE
- Maintain temporary identities across frames
- Associate detected PPE with the correct worker
- Separate external workers from confirmed cabin operators

Output

- Annotated video with worker and machine IDs
- PPE status for every tracked worker
- Relative danger-zone state and exposure duration
- Risk level with a human-readable reason
- Event log containing start time, end time and severity

Boundary of the claim

The system estimates relative risk in image space. An uncalibrated single camera cannot provide dependable real-world distance in metres.

Dataset inventory and sample images

MOCS
41,668 images, 13 moving-object classes. Used for workers and machinery.

SH17
8,099 images, 75,994 instances, 17 PPE and body-part classes.

CHV
1,330 selected images with person, vest and helmet-colour annotations.

SHWD
7,581 images with helmet-wearing and normal-head annotations.

Final allocation: MOCS trains the worker and machinery scene detector. SH17, CHV and SHWD train the PPE detector after class-label harmonisation.

Source: An et al. (2021); Ahmad and Rahimi (2024); Wang et al. (2021); official SHWD repository.

Dataset Characteristics

Dataset	Data type	Useful labels	Problem characteristics
MOCS	Real construction-site images	Workers and multiple moving-machine categories	Large scene variation, occlusion, scale differences and crowded backgrounds
SH17	Internet-sourced industrial images	Person, head, helmet, safety vest and other PPE/body parts	Severe class imbalance; many annotations occupy less than 1% of the image
CHV	Quality-filtered construction PPE images	Person, vest and four helmet colours	Construction-specific but small; helmet colours require label remapping
SHWD	Web and head-detection images	Helmet and normal head	Helmet-focused only; no vest, machinery, tracking or risk-event labels

Existing Limitations

- **Frame-only decisions:** a single incorrect detection can create an immediate warning.
- **No worker-specific PPE relationship:** detecting one helmet and two workers does not reveal who is protected.
- **No operator context:** a person inside a machinery cabin can be classified as an external worker in danger.
- **Static danger zones:** fixed rectangles or circles do not respond to observed machinery motion.
- **No exposure measurement:** brief entry and prolonged proximity receive the same treatment.
- **Weak cross-domain performance:** PPE accuracy drops when lighting, camera viewpoint or workplace type changes.
- **Incomplete datasets:** selected datasets contain object boxes, but not complete worker–PPE–machine risk-event annotations.
- **Unclear alert quality:** many studies report mAP but omit false alerts per hour and event-level recall.

Key Concepts Used in the Project

Concept	Role in SafeVision
Object detection	Predicts a class, confidence score and bounding box for each worker, machine or PPE item.
Training from random initialisation	Learns detector weights from the selected datasets without importing pretrained detector weights.
Label harmonisation	Maps different source labels into consistent scene and PPE class definitions.
IoU and non-maximum suppression	Measure box overlap and remove duplicate detections where the detector requires post-processing.
ByteTrack	Links detections across adjacent frames and assigns temporary worker and machine IDs.
PPE-to-worker association	Uses head and torso regions plus temporal voting to attach PPE to the most plausible worker.
Image-space danger zone	Represents relative proximity around machinery without claiming exact distance in metres.
Trajectory and dwell time	Estimate machine movement and measure how long a tracked worker remains exposed.
Temporal hysteresis	Requires persistent evidence before raising or clearing an alert.
Risk scoring	Combines proximity, motion, exposure duration and PPE state into an explainable severity level.

How the datasets will be used

Scene detector

Source: MOCS
Purpose: Detect workers and selected machinery classes.
Preparation:

- Retain classes required by the safety scenario
- Retain and rename the required MOCS classes consistently
- Keep provenance metadata for every image

PPE detector

Sources: SH17, CHV and SHWD
Purpose: Detect person/head, helmet and safety vest.
Preparation:

- Merge helmet-colour labels into one helmet class
- Map SHWD hat/person labels to helmet and uncovered head
- Balance minority PPE classes during sampling

Why two detectors

The scene and PPE datasets have different label spaces, object scales and visual distributions. Separate training avoids forcing unrelated annotations into one inconsistent model. Their outputs unite on each video frame during detection fusion.

Literature Review I

Study	Problem and dataset	Method and key result	Limitation for SafeVision
An et al. (2021), MOCS	Construction worker and machinery detection; 41,668 images from 174 sites	Benchmarked 15 deep detectors and established a construction-object benchmark	Frame-level objects only; no PPE association, tracking, proximity events or duration
Wang et al. (2021), CHV	Fast construction PPE detection; 1,330 selected images	YOLOv5x reported 86.55% mAP; YOLOv5s reported 52 FPS on the authors' GPU	Small dataset; detects appearance only and does not model machinery interaction
Ahmad and Rahimi (2024), SH17	PPE detection across manufacturing environments; 8,099 images and 17 classes	YOLOv9-e reached 70.9% mAP@0.5 and 48.7% mAP@0.5:0.95	Used transfer learning; cross-domain mAP@0.5 fell to about 58.9%; no temporal association

Literature Review II

Study	Problem and dataset	Method and key result	Limitation for SafeVision
Liu et al. (2025), SHWD	Small-helmet detection in heavy-machinery factories	YOLOv10-IYA reported 93.9% mAP@0.5 with 7.8 GFLOPs	Helmet/head only; missed detections remain; reported edge speed was about 0.9 FPS
Nath et al. (2020), Pictor-PPE	Real-time helmet and vest compliance	Compared three YOLOv3-based pipelines; unified model reported 72.3% mAP at 11 FPS	PPE compliance remains frame based; no operator state, machinery motion or exposure-time reasoning

Note: The 2025 SHWD paper reports dataset counts that conflict with the official SHWD repository. The official repository should be used for dataset statistics.

Research Gap

- Existing studies optimise detection metrics, but site deployment requires accurate safety-event decisions.
- Most studies do not measure whether detected PPE belongs to the correct worker.
- Frame-level methods cannot distinguish brief crossings from persistent exposure.
- A nearby cabin operator and an external pedestrian may receive the same proximity warning.
- Fixed image-space danger zones ignore observed machine movement.
- Existing datasets do not directly contain complete worker–PPE–machine event annotations.
- Cross-domain performance drops when models move between construction and manufacturing imagery.
- Few studies report false alerts per hour, event recall or alert delay.

Gap addressed by this project

An integrated video pipeline that converts two independent detectors into worker-specific, operator-aware and temporally stable risk events.

What is missing?

Missing from existing systems

- Verified PPE ownership for each worker
- Cabin-aware handling of machine operators
- Danger zones that react to observed movement
- Exposure duration and stable event boundaries
- Human-readable reasons for every risk alert
- Event-level false-alarm evaluation

Missing from public datasets

- Unified scene, PPE and temporal-risk labels
- Persistent worker and machine track IDs
- Cabin-operator annotations
- Ground-truth danger-zone entry intervals
- Calibrated real-world distance labels
- Enough site diversity for guaranteed deployment

Project response

Use large public object datasets for detector training and create a smaller, manually annotated video benchmark for system-level event evaluation.

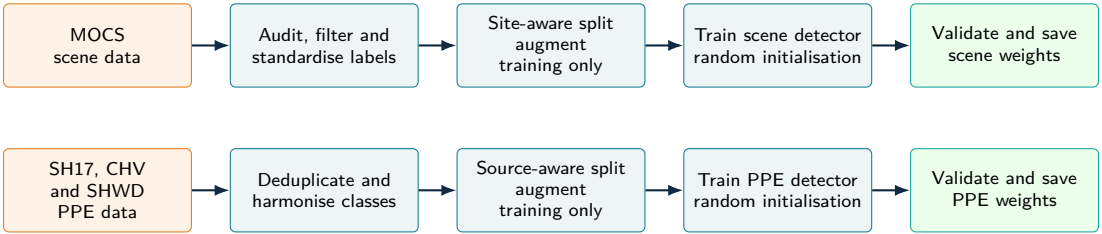
Project Objectives

1. Prepare the required MOCS classes and train a worker and machinery detector from random initialisation.
2. Build a harmonised PPE dataset from SH17, CHV and SHWD and train a second detector from random initialisation.
3. Track workers and machinery across frames and associate helmet and vest detections with the correct tracked worker.
4. distinguish confirmed cabin operators, external workers and uncertain cases without using facial recognition.
5. Estimate image-space machine motion and construct an adaptive danger zone around each tracked machine.
6. Generate explainable risk events using proximity, motion, PPE status and exposure duration.
7. Compare the complete system with a fixed-zone, single-frame baseline.

Proposed Solution / Methodology

1. Prepare MOCS for worker and machinery detection. Harmonise SH17, CHV and SHWD for person/head, helmet and safety-vest detection.
2. Train two specialised YOLO-style detectors from random initialisation and select checkpoints using validation performance.
3. Run both detectors on every video frame and fuse their outputs without merging the two training label spaces.
4. Use ByteTrack to maintain worker and machinery IDs, trajectories and exposure histories.
5. Associate helmet and vest detections with worker head and torso regions, then stabilise the result across several frames.
6. Estimate machinery movement, classify cabin/external/uncertain worker context and construct adaptive image-space danger zones.
7. Accumulate risk from zone occupancy, motion, PPE state and duration. Apply persistence and hysteresis before creating an alert.
8. Produce annotated video, explainable alerts and a structured event log.

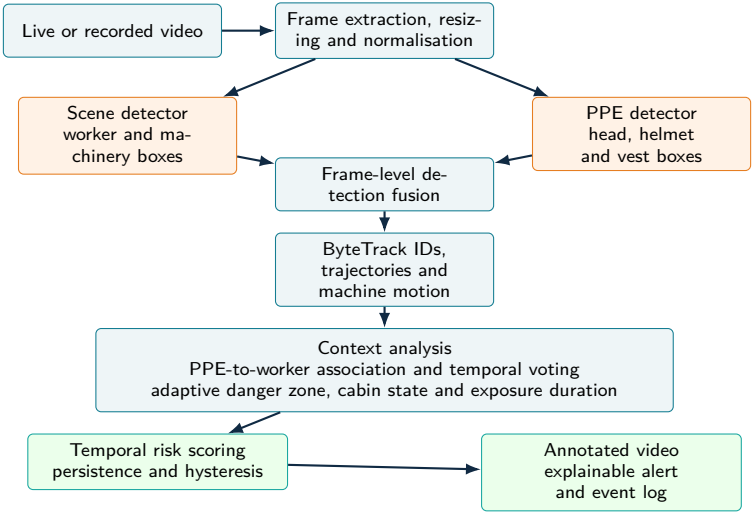
Model-development workflow



Training output

The saved scene and PPE weights enter the video-inference pipeline. The test sets remain untouched until final model evaluation.

Complete video-inference architecture



The two detectors unite after inference. Tracking and context logic convert their boxes into worker-specific, time-based safety events.

Preprocessing and augmentation pipeline

Data preparation

1. Verify labels, licences and corrupt files.
2. Preserve source and site/video provenance.
3. Remove exact and near duplicates.
4. Harmonise class names separately for each detector.
5. Split by site, source or video before augmentation.
6. Resize with letterboxing to the selected input size.

Training-only augmentation

- Horizontal flip where direction does not change the label
- Controlled brightness, contrast and colour changes
- Mild blur and image noise
- Scale, translation and limited crop
- Mosaic during the early training stage
- Class-aware sampling for rare PPE classes

Deep learning detection core

Model A: scene detector

- YOLO-style detector architecture
- Randomly initialised weights
- Trained on the selected MOCS worker and machinery classes
- Outputs class, confidence and bounding box for workers and machinery

Model B: PPE detector

- Same detector family with a separate classification head
- Randomly initialised weights
- Trained on harmonised SH17 and CHV data
- Outputs person/head, helmet and safety-vest detections

Training objective: optimise localisation, classification and distribution-focal losses using only training images. Select checkpoints using validation mAP and class-level recall.

Segmentation scope correction

The selected datasets mainly provide bounding boxes, not pixel masks. Therefore the feasible core is object detection. Instance segmentation becomes an optional extension only if the team creates reliable mask annotations.

Tracking, fusion and PPE association

1. Apply both detectors to the same frame.
2. Filter low-confidence and duplicate detections using class-specific thresholds and non-maximum suppression where required.
3. Send worker and machinery boxes to ByteTrack, which assigns temporary IDs and matches them across adjacent frames.
4. Assign each helmet to the worker whose upper-body region contains it with the strongest geometric agreement.
5. Assign each vest to the worker whose torso region contains it.
6. Use temporal voting across several frames to reduce one-frame PPE errors caused by occlusion.
7. Mark competing or weak associations as *uncertain*; do not force every PPE item onto a worker.

Meaning of identity

Worker 01 and Worker 02 are temporary tracking IDs within one video. The system does not recognise real people and does not need the same individuals to appear in the training datasets.

Operator-aware adaptive temporal risk engine

For tracked worker i at time t, the prototype computes

$$R_i(t) = G_i(t)\left[w_z Z_i(t) + w_m M_i(t) + w_\tau T_i(t) + w_p P_i(t)\right],$$

where Z represents danger-zone occupancy, M represents observed machinery motion or approach, T represents exposure duration, and P represents PPE non-compliance.

- $G_i(t)$ applies operator context. A confirmed cabin operator suppresses the external-pedestrian proximity alert.
- An uncertain cabin state never receives automatic safe status.
- The danger zone expands in the observed direction of machine movement.
- Risk accumulates during continued exposure and decays after the worker returns to a safe state.
- Hysteresis requires persistence before raising or clearing an alert.

Calibration

The weights and thresholds are engineering parameters, not legal safety distances. Tune them on annotated validation videos and report their sensitivity.

Novelty

Primary contribution

An operator-aware adaptive temporal risk engine that converts detections into worker-specific safety events.

- Two from-scratch detectors specialised for scene objects and small PPE objects, followed by explicit detection fusion.
- Worker-specific PPE association instead of scene-level PPE counting.
- Conservative separation of cabin operators, external workers and uncertain cases.
- Motion-adaptive image-space danger zones rather than one fixed radius.
- Exposure-time accumulation and temporal hysteresis to reduce unstable one-frame alerts.
- Explainable event output containing the worker ID, machine ID, detected causes, exposure duration and risk level.
- Ablation against a fixed-zone, single-frame baseline to measure whether the proposed context actually reduces false alerts.

Not claimed as novelty: the YOLO architecture, ByteTrack, or standard augmentations. The contribution lies in the integration and its event-level evaluation.

Dataset & Experimental Setup

Component	Data and configuration	Experimental rule
Scene detector	MOCS; selected worker and machinery classes	Use official splits when suitable; otherwise split by construction site to prevent scene leakage
PPE detector	SH17, CHV and SHWD; person/head, helmet and vest classes	Deduplicate across sources; merge helmet colours; keep uncovered-head evidence only when labels are compatible
Input processing	Proposed detector input of 640×640 with letterboxing	Apply normalisation consistently; augmentation belongs to the training split only
Training condition	YOLO-style detectors with random initialisation	Train models independently; use validation data for checkpoint and threshold selection
Tracking and context	Recorded construction videos with worker and machinery motion	ByteTrack requires detections but no task-specific model training
System benchmark	Approximately 30–50 short clips annotated with safe and unsafe interactions	Include cabin operators, external workers, brief crossings, persistent exposure, PPE variation and occlusion
Hardware	Available GPU for training and video demonstration	Report GPU model, batch size, inference resolution, FPS and end-to-end latency

Exact split counts and class totals must be reported only after the final downloaded datasets pass the audit and deduplication stage.

Dataset Preparation Status

Dataset	Current decision	Required preparation	Completion evidence
MOCS	Selected for the scene detector	Obtain authorised copy, retain required classes, inspect annotations and group splits by site	Final image/class counts and split manifest
SH17	Selected for the PPE detector	Download through official links, verify files, retain relevant PPE classes and measure imbalance	Cleaned annotations and class-frequency report
CHV	Selected for the PPE detector	Verify 1,330 images, merge helmet colours and inspect overlap with other sources	Duplicate report and harmonised labels
SHWD	Selected for helmet/head coverage	Use official repository statistics, verify labels and map hat/person carefully	Mapping document and manual sample audit
Video benchmark	Required for event evaluation	Collect permitted short clips and annotate track IDs, cabin state, zone intervals, PPE state and event boundaries	Versioned annotation export and test protocol

Work Plan

1. Audit dataset access, licences, annotations and class distributions.
2. Define the final scene and PPE class ontologies.
3. Remove duplicates and create leakage-resistant data splits.
4. Train and evaluate the scene detector from random initialisation.
5. Train and evaluate the PPE detector from random initialisation.
6. Integrate both detectors with ByteTrack and detection fusion.
7. Implement PPE-to-worker association and temporal voting.
8. Implement cabin-operator handling and adaptive machine danger zones.
9. Build the temporal risk engine and structured event log.
10. Annotate a small system-level video benchmark containing safe and unsafe interactions.
11. Compare the baseline and proposed systems using event-level metrics.
12. Package the demonstration, results, limitations and reproducibility files.

Expected Outcomes & Evaluation Plan

Component	Expected output	Evaluation criteria
Scene detector	Worker and machinery boxes with confidence scores	mAP@0.5, mAP@0.5:0.95, per-class precision, recall and confusion matrix
PPE detector	Person/head, helmet and vest boxes	Same detection metrics, with separate reporting for small and occluded PPE
Tracking	Stable worker and machine IDs within each video	IDF1, ID switches, track fragmentation and qualitative failure cases
PPE association	PPE status attached to each tracked worker	Association accuracy/F1 and incorrect-assignment rate
Context engine	Cabin-operator state and adaptive danger zone	Operator-state accuracy and cabin-related false alerts
Risk engine	Warning/critical events with reasons and duration	Event precision, recall, F1, missed-event rate, false alerts/hour and alert delay
Complete system	Annotated video and structured event log	End-to-end FPS, latency and resource usage

Software Stack and Reproducibility

Datasets

- MOCS: workers and construction machinery
- SH17: PPE and body-part annotations
- CHV: construction helmets and safety vests
- SHWD: helmet-wearing and uncovered-head data
- Small custom video evaluation set for temporal events

Software stack

- Python and PyTorch
- YOLO detector implementation with random initialisation
- OpenCV for video processing and geometry
- ByteTrack for multi-object tracking
- CVAT or Roboflow for annotation review
- NumPy, pandas and scikit-learn for evaluation
- Streamlit for the optional demonstration interface
- Git and GitHub for version control and reproducibility

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`https://github.com/njvisionpower/Safety-Helmet-Wearing-Dataset`

Ultralytics, "Ultralytics YOLO documentation and software repository."
`https://github.com/ultralytics/ultralytics`

Dataset pages and access conditions should be recorded in the final reproducibility appendix.

Thank You

Questions and discussion